

**GENFT 302 – Applied Mathematics
/ Apply Fundamental Algebra and
Trigonometry**

Date: 12 June 2024

Period: 08.30 am – 11.30 am



END OF TERM III EXAMINATIONS MARKING GUIDE

SECTOR: ICT & MULTIMEDIA

TRADES: Computer Systems and Architecture(CSA), Multimedia
Production (MMP), Software Development (SFD),
Networking and Internet Technologies (NIT).

TVET LEVEL:3

MODULE TITLE: APPLIED MATHEMATICS/ Apply
Fundamental Algebra and Trigonometry

MARKS: **CAMIS**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Three Sections (A, B). Follow the instructions given below, and answer the indicated questions for a total of 100 marks

Section **A**: Fifteen (15) questions, all **Compulsory** **55 marks**

Section **B**: Among the five (5) questions, attempt any three (3) **45 marks**

SECTION A: Attempt All questions**[55marks]**

- 01.** Given the parametric equation $f(x) = \frac{mx}{2m+3}$, state the condition for which the parametric linear equation with parameter m has the solution. **[2marks]**

Answer:

The condition for which the parametric linear equation $f(x) = \frac{2mx}{m+3}$ has the solution is where $2m + 3 \neq 0$ **/ 1mark**

$$m \neq -3/2 \quad \textbf{/ 1mark}$$

References: LO1.Solve linear and quadratic equations or inequality. IC 1.2. Discussion on parameter from a given linear equation. Curriculum level 3 year 2023, page 2). Cognitive level: Remembering.

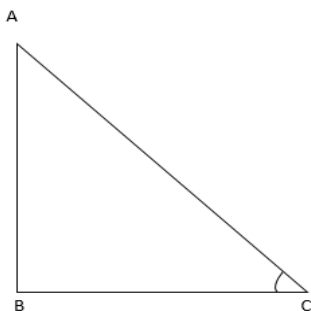
- 02.** Define the following terms **[3marks]**
- a. Light angle
 - b. Obtuse angle
 - c. Reflex angle

Answers

- a. Light angle: An angle exactly with 90 degrees **/ 1mark**
- b. Obtuse angle: An angle whose measure is greater than 90^0 and less than 180^0 **/ 1mark**
- c. Reflex angle: An angle whose measure is greater than 180^0 and less than 360 degree **/ 1mark**

References: LUs: 3. Apply fundamentals of trigonometry, IC.3.1. Description of angles, Curriculum 2023, page3). Cognitive level: remembering.

- 03.** Consider the given triangle and determine the $\cos \hat{c}$, $\sin \hat{c}$ and $\tan \hat{c}$. **[3marks]**



Answers

$$\cos \hat{c} = \frac{BC}{AC}, \text{ /1mark } \sin \hat{c} = \frac{AB}{AC}, \text{ /1mark } \tan \hat{c} = \frac{AB}{BC} \text{ /1mark}$$

References: LUs 3. Apply fundamentals of trigonometry, IC.3.4

Solving trigonometric equation, Curriculum version 2023, page4).

Cognitive level: understanding.

04. If matrix $A = \begin{pmatrix} 7 & 1 & -3 \\ 2 & 4 & 6 \\ 0 & z & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 7 & 1 & -3 \\ 2 & x & 6 \\ 0 & 3 & y \end{pmatrix}$ are equal, find the value

of x , y and z

[3marks]

Answers

$$x = 4 \text{ /1mark}, y = -1 \text{ /1mark}, z = 3 \text{ /1mark}$$

References (LUs 2: Apply matrices in solving simultaneous linear equations, IC.2.2. Computation of determinant, Curriculum version 2023, page3) Cognitive level: remembering.

05. Define the following terms:

- a. Upper triangular matrixes **[1mark]**
- b. Diagonal matrix **[1mark]**
- c. Scalar matrix **[1mark]**
- d. Identity matrix **[1mark]**
- e. Lower triangular matrix **[1mark]**

Answers

- a. An upper triangular matrix, is a matrix whose all elements located below the leading diagonal are zeros **/1mark**

- b. A diagonal matrix, , is a matrix whose all the elements above and below the leading diagonal are zeros. /1mark
- c. A scalar matrix is a diagonal matrix in which the leading diagonal elements are equal. /1mark
- d. An identity matrix (denoted by I) is a diagonal matrix in which the leading diagonal elements are equal to 1. /1mark
- e. A lower triangular matrix, , is a matrix whose all the elements above the leading diagonal are zeros. /1mark

References: LUs 2. Apply matrices in solving simultaneous linear equations, IC.2.1. Description of matrices properties, Curriculum version 2023, page3). Cognitive level: remembering.

06. Find the determinant of this given matrix $A = \begin{pmatrix} 17 & 1 & -23 \\ 0 & 0 & 6 \\ 0 & 0 & -1 \end{pmatrix}$

Justify the answer [3marks]

Answers

$$\det(A) = 17 \begin{vmatrix} 0 & 6 \\ 0 & -1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 6 \\ 0 & -1 \end{vmatrix} - 23 \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} \quad /1mark$$

$$17(0) - 1(0) - 13(0) = 0 \quad /1mark$$

This matrix is singular matrix /1mark

References (LUs 2: Apply matrices in solving simultaneous linear equations, IC.2.2. Computation of determinant, Curriculum version 2023, page3). Cognitive level: understanding.

07. Without using a calculator, find the value of ϕ if $\begin{cases} \sin \phi = \frac{\sqrt{3}}{2} \\ \cos \phi = -\frac{1}{2} \end{cases}$ [2marks]

Answers

The Value of $\theta = 120$ or $\frac{2\pi}{3}$ /2marks

References LUs 3: Apply fundamentals of trigonometry, IC.3.2. Determination of trigonometric ratios, Curriculum version 2023, page4). Cognitive level: remembering.

08. Considering the following matrices, which ones are square matrices.

[3marks]

$$A = \begin{pmatrix} 3 \\ 8 \\ 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 5 \\ 2 & 1 \\ 5 & 4 \end{pmatrix}, C = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}, D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, E = \begin{pmatrix} 1 & p \\ 6 & 2 \end{pmatrix}, F = (1 \ 3)$$

Answers

$$C = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}, \text{ /1mark } D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \text{ /1mark } E = \begin{pmatrix} 1 & p \\ 6 & 2 \end{pmatrix} \text{ /1mark}$$

References (LUs 2: Apply matrices in solving simultaneous linear equations, IC.2.1 Description of matrices properties, Curriculum version 2023, page3) Cognitive level: remembering.

09. Solving the following equation $\begin{cases} y = x^2 \\ y = x + 6 \end{cases}$ **[5marks]**

Answers

$$\begin{cases} y = x^2 \text{ (eq1)} \\ y = x + 6 \text{ (eq2)} \end{cases}$$

Insert equation 1 in 2

$$x^2 = x + 6 \Rightarrow x^2 + x - 6 = 0$$

/0.5marks

$$x^2 - 3x + 2x - 6 = 0$$

/0.5marks

$$(x^2 - 3x) + (2x - 6) = 0$$

$$x(x - 3) + 2(x - 3) = 0$$

/1mark

$$(x + 2)(x - 3) = 0 \quad x = -2 \text{ or } x = 3$$

/1mark

$$\text{When } x = -2, y = 4 \text{ or } x = 3, y = 9$$

/1mark

$$S = \{(-2, 4), (3, 9)\}$$

/1mark

References (LUs 1: Solve algebraically or graphically linear and quadratic equations or inequalities, IC.1.4. Solving a quadratic equation and inequality, Curriculum version 2023, page2). Cognitive level: applying.

10. Compute the determinant of matrix $A = \begin{pmatrix} 2 & 5 \\ -5 & 0 \end{pmatrix}$ **[2marks]**

Answers

$$\begin{vmatrix} 2 & 5 \\ -4 & 0 \end{vmatrix} = 2 \times 0 - (-4 \times 5) \\ = 0 + 20 \\ = 20$$

/1mark

/1mark

References LO2 (computation of determinant), IC.2.1. Description of matrices properties, Curriculum level 3 year 2023, page 4). Cognitive level: understanding.

11. Solve the simultaneous linear equation $\begin{cases} 3x - y = 6 \\ 2x + 4y = 4 \end{cases}$ **[5marks]**

Answer

By using comparison

$$-y = 6 - 3x, y = 3x - 6 \text{ eq3}$$

/0.5marks

$$4y = 4 - 2x \quad y = 1 - \frac{1}{2}x \text{ eq4}$$

/0.5marks

Equality of equation 1 and equation 2

$$3x - 6 = 1 - \frac{1}{2}x \Rightarrow 3x + \frac{1}{2}x = 6 + 1$$

/1mark

$$\frac{6x+x}{2} = \frac{14}{2} \Leftrightarrow 6x + x = 14$$

/1mark

$$7x = 14 \Rightarrow x = 2 \quad x \text{ in eq4} \quad y = 1 - \frac{1}{2} * 2 \Rightarrow y = 0$$

/1mark

$$S = \{(2,0)\}$$

/1mark

References (LUs 1: Solve algebraically or graphically linear and quadratic equations or inequalities, IC.1.3. Solving two simultaneous linear equations, Curriculum version 2023, page2). Cognitive level: Understanding.

12. Using the following simultaneous linear equation: $\begin{cases} 3x - y = 0 \\ x - 2(x - y) = 1 \end{cases}$

a. Write this system of equation in the $AX = B$ form **[2marks]**

b. Find the determinant of matrix A **[2marks]**

Answer

a. $\begin{cases} 3x - y = 0 \\ x - 2(x - y) = 1 \end{cases}$

$$= \begin{cases} 3x - y = 0 \\ -x + 2y = 1 \end{cases}$$

/1mark

The system of equation in the form matrix multiplication $AX = B$ is

$$\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad /1\text{mark}$$

b. Det. A = $\begin{vmatrix} 3 & -1 \\ -1 & 2 \end{vmatrix} = 3 \times 2 - (-1 \times -1)$ /1mark

$$= 6 - 1$$

$$= 5 \quad /1\text{mark}$$

References (LUs 2: Apply matrices in solving simultaneous linear equations, IC.2.2. Computation of determinant, Curriculum version 2023, page3). Cognitive level: Analyzing.

13. Solve the following equation and inequality [5marks]

a. $x^2 - 4x + 3 = 0$

b. $6x^2 - 5x + 1 \leq 0$

Answers

a. $D = (-4)^2 - 4(1)(3) = 4$ /1mark

$$x_1 = \frac{4+\sqrt{4}}{2} = 3, x_2 = \frac{4-\sqrt{4}}{2} = 1$$
 /1mark

$$S = \{(1, 3)\}$$

b. Equivalent equation $6x^2 - 5x + 1 = 0$

$$D = (-5)^2 - 4(6)(1) = 1$$

$$x_1 = \frac{5+\sqrt{1}}{6*2} = \frac{1}{2}, x_2 = \frac{5-\sqrt{1}}{6*2} = \frac{3}{4}$$
 /1mark

x	$-\infty$	$\frac{1}{2}$	$\frac{3}{4}$	$+\infty$
<i>factor</i>				
$6x^2 - 5x + 1$	+++++	-----	+++++	

/1mark

$$S, x \in \left[\left(\frac{1}{2}, \frac{3}{4}\right)\right]$$
 /1mark

References: LUs 1: Solve algebraically or graphically linear and quadratic equations or inequalities, IC.1.1. Solving linear equations and inequalities, Curriculum version 2023, page2). Cognitive level: applying.

14. Solve the equation $\cos 2x + \cos 4x = 0$ [5marks]

Answer

$$-2 \sin \frac{2x+4x}{2} \sin \frac{2x-4x}{2} \quad /1\text{mark}$$

$$-2 \sin \frac{6x}{2} \sin \frac{-2x}{2} \quad /1\text{mark}$$

$$2 \sin 3x \sin x = 0$$

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$$\sin 3x \sin x = 0 \Rightarrow \sin 3x = 0 \text{ and } \sin x = 0$$

$$\sin 3x = \sin \left\{ \frac{k\pi}{\pi + 2k\pi} \right\} \text{ or } \sin x = \sin \left\{ \frac{k\pi}{\pi + 2k\pi} \right\} \quad / 1 \text{ mark}$$

$$3x = \left\{ \frac{k\pi}{\pi + 2k\pi} \right\} \text{ or } x = \left\{ \frac{k\pi}{\pi + 2k\pi} \right\} \quad / 1 \text{ mark}$$

$$x = \left\{ \frac{\frac{1}{3}k\pi}{\frac{\pi}{3} + \frac{2}{3}k\pi} \right\} \quad GS = \left\{ k\pi, \frac{\pi}{3} + 2k\pi, \pi + 2k\pi \right\} \quad / 1 \text{ mark}$$

References: LUs 3. Apply fundamentals of trigonometry, IC.3.4. Solving trigonometric equations, Curriculum version 2023, page4). Cognitive level: applying.

15. Given that $f(x) = x^2 - 4x + 3$

a) Write the name of the given equation; [1mark].

b) Find x -intercepts; [4marks].

Answers:

a) $f(x) = x^2 - 4x + 3$ is a quadratic function. / 1 mark,

b) x -intercepts, let $f(x) = 0$ / 0.5 marks. $x^2 - 4x + 3 = 0$ implies that $(x - 3)(x - 1) = 0$ / 0.5 marks either $x - 3 = 0$, $x = 3$ / 0.5 marks. $x - 1 = 0$ $x = 1$ / 0.5 marks. thus x -intercept at $(1, 0)$ / 1 mark and $(3, 0)$ / 1 mark.

References (LUs 1: Solve algebraically or graphically linear and quadratic equations or inequalities, IC.1.4. Solving a quadratic equation and inequality, Curriculum version 2023, page2) (Reference: Evaluating)

SECTION B: Attempt any Three (3) questions [45 marks]

16.

a. Simplify $\left(\frac{1}{\tan x} + \frac{1}{\cot x} \right) \sin x \cos x$ [7marks]

b. Verify if $\frac{\cos^2 x}{1 + \tan^2 x} - \frac{\sin^2 x}{1 + \tan^2 x} = 1 - 2\sin^2 x$ [8marks]

Answers

$$\begin{aligned}
 \text{a. } \left(\frac{1}{\tan x} + \frac{1}{\cot x}\right) \sin x \cos x &= \left(\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}\right) \sin x \cos x & /2\text{marks} \\
 &= \left(\frac{\cos x \cos x + \sin x \sin x}{\sin x \cos x}\right) \sin x \cos x & /1\text{mark} \\
 &= \cos x \cos x + \sin x \sin x & /2\text{marks} \\
 &= \cos^2 x + \sin^2 x & /1\text{mark} \\
 &= 1 & /1\text{mark}
 \end{aligned}$$

$$\begin{aligned}
 \text{b. } \frac{\cos^2 x}{1+\tan^2 x} - \frac{\sin^2 x}{1+\cot^2 x} &= \frac{\cos^2 x}{1+\frac{\sin^2 x}{\cos^2 x}} - \frac{\sin^2 x}{1+\frac{\cos^2 x}{\sin^2 x}} & /2\text{marks} \\
 &= \frac{\cos^2 x}{\frac{\cos^2 x + \sin^2 x}{\cos^2 x}} - \frac{\sin^2 x}{\frac{\sin^2 x + \cos^2 x}{\sin^2 x}} & /2\text{marks} \\
 &= \frac{\cos^2 x}{1} - \frac{\sin^2 x}{1} & /1\text{mark} \\
 &= \cos^2 x - \sin^2 x \\
 &= (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) & /1\text{mark} \\
 &= (\cos^2 x - \sin^2 x) \cdot 1 \\
 &= \cos^2 x - \sin^2 x \\
 &= \cos^2 x - \sin^2 x = 1 - \sin^2 x - \sin^2 x & /1\text{mark} \\
 &= 1 - 2\sin^2 x & /1\text{mark} \\
 &\text{Expression proved}
 \end{aligned}$$

References. LUs3: Apply fundamentals of trigonometry, IC.3.3.

Description of trigonometric identities, Curriculum version 2023, page4). Cognitive level: Evaluating.

17.

- a. The sum of two numbers is 25. One of the numbers exceeds the other by 9. Find the numbers. **[7marks]**

b. If $C = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & 3 & -1 \\ -1 & 2 & 2 \\ 0 & 1 & -1 \end{pmatrix}$, what do you conclude about $|C \times D|$ and $|C| \times |D|$ **[8marks]**

Answers

- a. Let x be the first number and y be the second number **/1mark**

$$\begin{cases} x + y = 25 \\ x + 9 = y \end{cases} \Rightarrow \begin{cases} x + y = 25 \\ x - y = -9 \end{cases} \quad \text{/1mark} \text{ By adding the equation}$$

$$2x = 25 - 9 \quad \text{/1mark} \quad \text{/1mark}$$

$$2x = 16 \quad x = \frac{16}{2} \quad x = 8 \quad \text{/1mark} \quad y = 25 - x, y = 25 - 8, y = 17 \quad \text{/1mark}$$

There for those number are **8** and **17** /1mark

$$\text{b. } C \times D = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 & -1 \\ -1 & 2 & 2 \\ 0 & 1 & -1 \end{pmatrix} =$$

$$\begin{pmatrix} 1+0+0 & 3+0-1 & -1+0+1 \\ 0-2+0 & 0+4+1 & 0+4-1 \\ 0-2+0 & 0+4+3 & 0+4-3 \end{pmatrix} \quad \text{/1mark}$$

$$\begin{pmatrix} 1 & 2 & 0 \\ -2 & 5 & 3 \\ -2 & 7 & 1 \end{pmatrix} \quad \text{/1mark}$$

$$\text{Det}(C \times D) = 1 \begin{vmatrix} 5 & 3 \\ 7 & 1 \end{vmatrix} - 2 \begin{vmatrix} -2 & 3 \\ -2 & 1 \end{vmatrix} = (5 - 21) - 2(-2 + 6)$$

/1mark

$$= -16 - 8 = -24$$

/1mark

$$\text{Det}(C) = 1 \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} - 1 \begin{vmatrix} 0 & 2 \\ 0 & 2 \end{vmatrix} = (6 - 2) - (0 - 0) = 4$$

/1mark

$$D = \begin{pmatrix} 1 & 3 & -1 \\ -1 & 2 & 2 \\ 0 & 1 & -1 \end{pmatrix}$$

/1mark

$$\text{Det}(D) = 1 \begin{vmatrix} 2 & 2 \\ 1 & -1 \end{vmatrix} - 3 \begin{vmatrix} -1 & 2 \\ 0 & -1 \end{vmatrix} - 1 \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix}$$

$$= (-2 - 2) - 3(1 - 0) - 1(-1 - 0)$$

/1mark

$$= -4 - 3 + 1 = -6$$

$$\text{Det}(C) \times \text{Det}(D) = -6 \times 4 = -24$$

/0.5marks

There for $|C \times D|$ and $|C| \times |D|$ or $\text{Det}(C \times D) = \text{Det}(C) \times \text{Det}(D)$

are equal

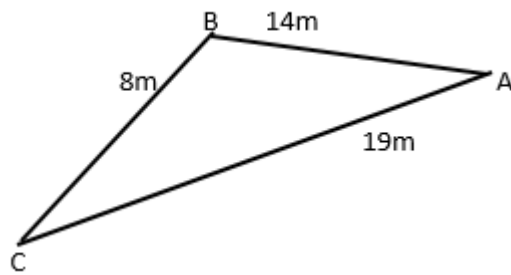
/0.5marks

References: LUs2: Apply matrices in solving simultaneous linear equations, IC.2.3. Solving simultaneous linear equations, Curriculum version 2023, page3). Cognitive level: Applying.

- 18.** A triangular shape site has the measurements of 8m, 14m and 19m respectively at its sides. Determine the angles of that site. [15marks]

Answers

The shape of that site



/3mark

$$\cos \hat{B} = \frac{8^2 + 14^2 - 19^2}{2(8)(14)} = \frac{64 + 196 - 361}{224} = \frac{-101}{224} = -0.45089$$

/1mark

/1mark

/1mark

$$\hat{B} = \cos^{-1} -0.45089 = 116.8^{\circ}$$

/1mark

$$\cos \hat{A} = \frac{19^2 + 14^2 - 8^2}{2(19)(14)} = \frac{361 + 196 - 64}{532} = \frac{493}{532} = 0.9267$$

/1mark

/1mark

/1mark

$$\hat{A} = \cos^{-1} 0.9267 = 22.07^{\circ}$$

/1mark

$$\cos \hat{C} = \frac{8^2 + 19^2 - 14^2}{2(8)(19)} = \frac{64 + 361 - 196}{304} = \frac{229}{304} = 0.7533$$

/1mark

/1mark

/1mark

$$\hat{C} = \cos^{-1} 0.7533 = 41.12^{\circ}$$

/1mark

The angle of that sites are 22.07° , 116.8° , 41.12°

References: LUs 3: Apply fundamentals of trigonometry, IC.3.5. Solving a given triangle, Curriculum version 2023, page5). Cognitive level: Analyzing.

19. Solve graphically the following equation and inequality

a. $2x + 5y \geq 6$ and shade the region which contains the solution. **[7marks]**

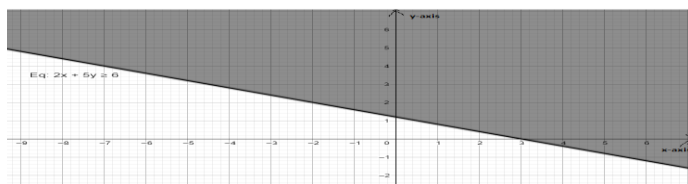
b. $\begin{cases} 3x - 4y = 1 \\ x - 3y = 2 \end{cases}$ **[8marks]**

Answers

a. Graphical solving

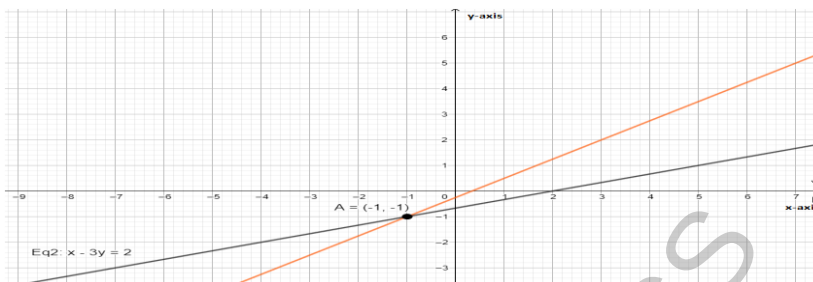
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$2x + 5y \geq 6$ and shade the wanted solution



/7marks

b. Graphical solving $\begin{cases} 3x - 4y = 1 \\ x - 3y = 2 \end{cases}$



/7marks

$$S = \{(-1, -1)\}$$

/1mark

References (LUs 1: Solve algebraically or graphically linear and quadratic equations or inequalities, IC.1.4. Solving a quadratic equation and inequality, Curriculum version 2023, page2).

Cognitive level: creating

- 20.** Find the exact values of the five trigonometric functions of an angle ϕ , in standard position given that $\cos \phi = \frac{3}{5}$ and the angle ϕ is in first quadrant

[15marks]

Answers

We know that $\sin^2 \theta + \cos^2 \theta = 1$

/1mark

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

/1mark

$$= \sqrt{1 - \left(\frac{3}{5}\right)^2}$$

/1mark

$$= \sqrt{1 - \frac{9}{25}}$$

$$= \sqrt{\frac{25-9}{25}}$$

$$= \sqrt{\frac{16}{25}}$$

$$= \frac{4}{5}$$

/1 mark

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\frac{4}{5}}{\frac{3}{5}} = \frac{4}{3}$$

/2 marks

$$\cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\frac{3}{5}}{\frac{4}{5}} = \frac{3}{4}$$

/1 mark

$$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{3}{5}}$$

/2 marks

$$= \frac{5}{3}$$

/1 mark

$$\csc \theta = \frac{1}{\sin \theta} = \frac{1}{\frac{4}{5}}$$

1/ mark /1 mark

$$= \frac{5}{4}$$

/1 mark

References (LUs 3: Apply fundamentals of trigonometry, IC.3.2.

Determination of trigonometric ratios, Curriculum version 2023, page4).

END OF ASSESSMENT!